

Homework2

Note: Submit your work (**upload the .java source code files ONLY, not the compiled .class files!**) through the “Homework2” link on **Codegrade**. You may submit an unlimited number of times; we will only grade the last/latest submission attempt, but be sure to **attach all of your files to each submission attempt**. Be sure to include your name and ID number in a comment at the beginning of each file that you submit.

Due: Tuesday, November 11, 11:59pm **Total: 25 points (5 points per problem)**

Submission Instructions: Name your java classes for this assignment as:

Problem1: Pyramid.java
Problem2: Interests.java
Problem3: LongestCommonPrefix.java
Problem4: PerfectNumber.java
Problem5: ArmstrongNumbers.java

1. Pyramid.java: (Printing numbers in a pyramid pattern) Write down a program in Java with a nested for loop that prints the following output (powers of 2) for any number of lines:

Here is a sample run:

Enter the number of lines: 8

Output:

```

      1
    1 2 1
  1 2 4 2 1
1 2 4 8 4 2 1
 1 2 4 8 16 8 4 2 1
1 2 4 8 16 32 16 8 4 2 1
1 2 4 8 16 32 64 32 16 8 4 2 1
1 2 4 8 16 32 64 128 64 32 16 8 4 2 1
```

2. Interests.java: (Financial application: comparing loans with various interest rates) Write a program that lets the user enter the double loan amount and loan period in number of years (int) and displays the monthly and total payments for each interest rate starting from 5% to 8%, with an increment of 1/8.

Here is a sample run:

Loan amount: 10000.00

Number of years: 5

Interest Rate	Monthly Payment	Total Payment
5.000%	188.71	11322.74
5.125%	189.28	11357.13
5.250%	189.85	11391.59

...		
7.875%	202.16	12129.97
8.000%	202.76	12165.84

3. LongestCommonPrefix.java: Write a program in Java that prompts the user to enter two strings and display the largest common prefix of the two strings. If there are no common prefix between the two entered strings display a message which tells the user that the two string doesn't have a common prefix.

Here are some sample runs:

Enter the first string: Atlanta
Enter the second string: Macon
Atlanta and Macon have no common prefix.

Enter the first string: Welcome to Java
Enter the second string: Welcome to programming
The common prefix is: Welcome to
Note: The prefix actually includes the space after 'to' as well

Enter the first string: I love coffee
Enter the second string: I love Java
The common prefix is I love

4. PerfectNumber.java: A positive integer is called a perfect number if it is equal to the sum of all of its positive divisors, excluding itself. For example, 6 is the first perfect number because $6 = 3 + 2 + 1$. The next is $28 = 14 + 7 + 4 + 2 + 1$. Write a program that asks the user for an upper limit and prints all the perfect numbers up to that limit.

Here is a sample run:
Enter the upper limit: 10000
The perfect numbers below 10000 are: 6 28 496 8128.

5. ArmstrongNumbers.java: An Armstrong number is an n-digit integer such that the sum of the n^{th} power of its digits is equal to the number itself. For example, 371 is an Armstrong number because $3^3 + 7^3 + 1^3 = 371$ (371 is a 3-digit number). 8208 is an Armstrong number because $8^4 + 2^4 + 0^4 + 8^4 = 8208$ (8208 is a 4-digit number). Write a program that asks the user for a lower limit and an upper limit and prints all the Armstrong numbers up to that limit.

Here are some sample runs:

Enter the lower limit: 10

Enter the upper limit: 1000

The Armstrong numbers between 10 and 1000 are: 153 370 371 407

Enter the lower limit: 8000

Enter the upper limit: 20000

The Armstrong numbers between 8000 and 20000 are: 8208 9474

Enter the lower limit: 25000

Enter the upper limit: 100000

The Armstrong numbers between 25000 and 100000 are: 54748 92727 93084